

Brazil E-Commerce Sales Analysis

Data Analytics Portfolio Project

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Tools Used:

- **SQL**
 - **Power BI**
 - **Power Query**
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Project Summary:

This project analyzes a Brazilian e-commerce dataset to understand revenue performance, customer activity, product category performance, delivery efficiency, and customer satisfaction. The goal of the analysis is to identify which products and regions drive the most revenue, and how revenue changes over time. SQL was used to clean, join, and analyze the data, while Power BI was used to build an interactive dashboard displaying key business KPIs and trends.

Business Problem:

The company lacked a centralized reporting solution for monitoring revenue, orders, customers, product categories, regional performance, delivery times, and customer reviews.

The analysis was designed to answer the following business questions:

- How much revenue is the company generating?
 - How are revenue and orders changing over time?
 - Which states contribute the most revenue?
 - Which product categories generate the most sales?
 - Which regions have the highest average order values?
 - How does delivery time relate to customer review scores?
 - Which areas represent the strongest growth or operational-improvement opportunities?
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Project Objectives:

Primary Objective

Develop an interactive business intelligence dashboard to analyze e-commerce sales performance, customer behavior, and operational efficiency, enabling data-driven decision-making.

Specific Objectives

1. Monitor overall business performance
 - Track Total Revenue, Total Orders, Total Customers, and Average Order Value.
2. Identify sales trends
 - Analyze revenue performance over time to detect growth patterns and seasonal fluctuations.
3. Evaluate product performance
 - Determine which product categories generate the most revenue and contribute most to business success.
4. Analyze geographic performance
 - Compare revenue across states to identify high-performing and underperforming regions.
5. Assess customer satisfaction
 - Examine review scores and their distribution to understand the customer experience.
6. Evaluate delivery performance
 - Investigate the relationship between delivery times and customer review scores.

Dataset Overview:

Dataset: Olist Brazilian E-Commerce Dataset

Source: Kaggle.com

Time Period: September 2016 – October 2018

Description:

The dataset contains information on customer orders, payments, products, reviews, and delivery performance from a Brazilian e-commerce marketplace.

Key Business Questions

1. How is overall revenue performing?
2. Which product categories generate the most revenue?
3. Which state contributes the most sales?
4. Who are the highest-value customers?
5. How satisfied are customers?
6. Does delivery speed impact review scores?
7. How has revenue changed over time?

KPIs	Value	Description
Total revenue:	\$16,008,872.12	Total Sales generated
Total Customers:	96,096	Number of unique customers
Total Orders:	99,441	Number of unique orders
Average Order Value:	\$160.99	Average revenue per order
Average Review Score:	4.1 Stars (out of 5)	Average review score
Average Delivery Time:	12.4 Days	Average time from purchase to delivery

SQL Analysis

Key SQL Queries: 6 examples

Total Revenue

```
SELECT
  SUM(payment_value) AS revenue
FROM order_payments
```

Average Order Value

```
SELECT
  SUM(payment_value) / COUNT(DISTINCT Order_ID) AS AvgValuePerOrder
FROM clean_payment
```

Average Review Score

```
SELECT
  ROUND
  (
    SUM(review_score) * 1.0 / COUNT(DISTINCT Order_ID),
    1
  ) AS AvgReviewScore
FROM clean_review
```

Top Categories by Revenue

```
SELECT
  p.product_category_name,
  SUM (oi.price + oi.freight_value) AS Revenue
FROM order_items oi
JOIN products p
  ON oi.product_id = p.product_id
GROUP BY p.product_category_name
ORDER BY Revenue DESC
```

*NOTE: Revenue by category requires a calculation of revenue using (price * freight_value)*

Revenue Month over Month

```
WITH TotalRevenue AS (
  SELECT
    FORMAT(o.order_purchase_timestamp, 'yyyy-MM') AS OrderMonth,
    SUM(op.payment_value) AS revenue
  FROM order_payments op
  JOIN orders o
    ON op.order_id = o.order_id
  GROUP BY
    FORMAT(o.order_purchase_timestamp, 'yyyy-MM')
)

SELECT
  OrderMonth,
  revenue,

  ROUND (
    (revenue - LAG(revenue) OVER (ORDER BY OrderMonth))
    * 100 /
    LAG(revenue) OVER (ORDER BY OrderMonth),
    2
  ) AS PercentageChange

FROM TotalRevenue
ORDER BY OrderMonth
```

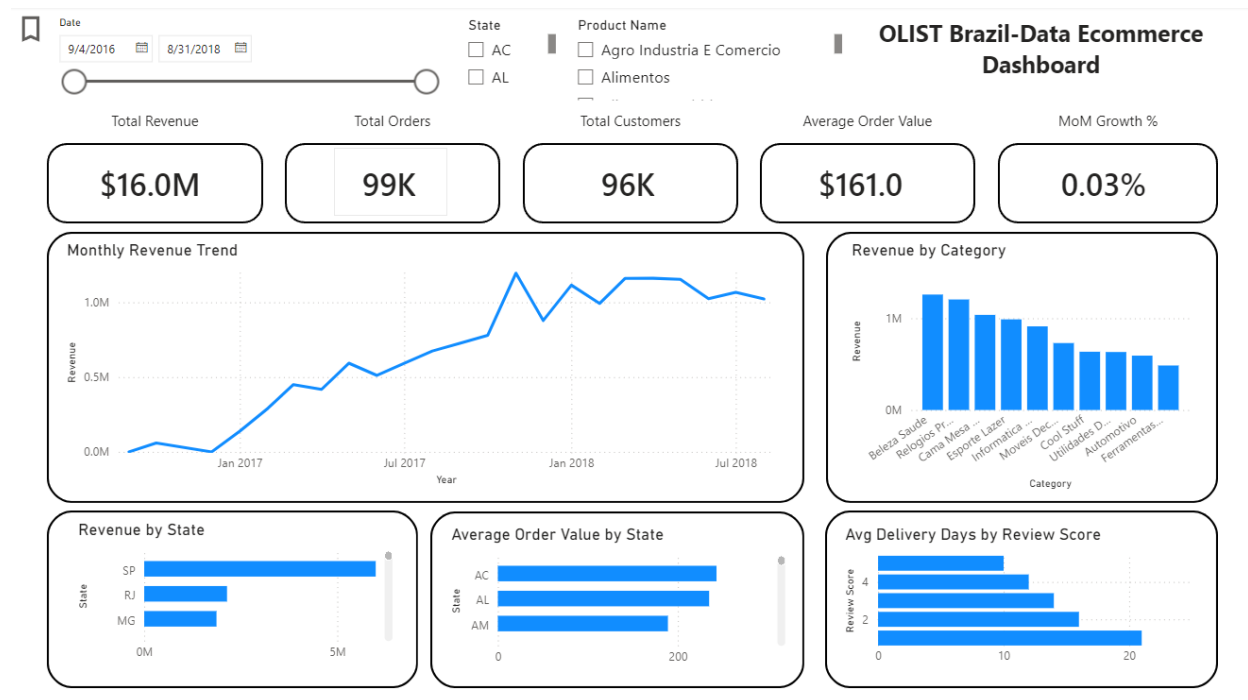
Average Delivery time by Review Score

```

WITH DeliveryReview AS
(
SELECT
    review_score,
    ROUND(
        AVG(
            DATEDIFF(
                day,
                o.order_purchase_timestamp,
                o.order_delivered_customer_date
            )
        ) * 1.0
    ), 1
) AS AvgDeliveryDays
FROM orders o
JOIN order_review ot
    ON o.order_id = ot.order_id
WHERE order_delivered_customer_date IS NOT NULL
GROUP BY review_score
)
SELECT *
FROM DeliveryReview

```

Power BI Dashboard:



KPI's:

1. Total Revenue

Measures the total sales generated by the business during the selected period.

2. Total Orders

Measures overall sales volume and customer purchasing activity.

3. Total Customers

Tracks the size of the customer base and customer reach.

4. Average Order Value

Measures the average amount spent per order.

5. Average Delivery by Review Score

Measures the relation between poor review score and average delivery time.

Charts:

1. Revenue growth month over month

Track revenue trends and identify periods of growth or decline.

2. Top 10 products by revenue

Identify the products, generating the most revenue and contributing most to overall business performance.

3. Revenue by state

Analyze geographic sales performance and identify high-performing regions.

4. Average Order Value by State

Identify which state the largest average order value.

5. Delivery Time vs. Review Score

Identify if longer average delivery times have a correlation to lower review scores.

Sliders:

1. Date Range Selector

2. State Selector

3. Product selector

Insights:

State:

1. São Paulo (SP) generated 37% of total company revenue, making it the largest revenue-producing state. However, it also had one of the lowest average order values at \$143.60 per order, suggesting that its revenue is driven primarily by high order volume rather than larger individual purchases.
2. Acre (AC) ranked 25th in total revenue (\$19,980) but 2nd in average order value (\$242.90). This indicates that while order volume is low, customers tend to spend significantly more per order than in most other states. Acre may warrant further investigation as a potential growth market.

Product/Category:

3. The top 10 product categories accounted for more than 60% of total revenue, indicating that a relatively small number of categories drive the majority of sales. This concentration may create both growth opportunities and dependency risk.

Customer Satisfaction:

4. Lower review scores were associated with significantly longer delivery times, suggesting delivery performance is a major driver of customer satisfaction.
5. One-star ratings makes up 11% of all ratings and is the third most selected review score behind 4 stars (2nd) and 5 stars (1st). The distribution should be closer to a descending order model instead.

Recommendations:

1. Prioritize São Paulo for customer retention and logistics optimization because it is the company's largest revenue market.
2. Investigate Acre further to determine whether targeted marketing could increase order volume while maintaining its high average order value.
3. Protect and expand the top-performing product categories but monitor dependency risk by identifying emerging categories with growth potential.
4. Improve delivery speed and reliability, especially for orders at risk of long delivery times, because delayed deliveries appear linked to lower review scores.
5. Analyze one-star reviews in more detail to identify recurring complaints related to delivery, product quality, or customer service.

Conclusion:

The analysis found that revenue is heavily concentrated in São Paulo and a limited number of product categories, while certain smaller markets such as Acre demonstrate high average order values.

It also found that longer delivery times are associated with lower customer review scores, making logistics performance an important area for operational improvement.

The strongest immediate business priorities are protecting revenue in São Paulo, improving delivery reliability, investigating high-value regional opportunities, and reducing dependency on a narrow group of product categories.